

HITL Maturity Ratios

PCF 4.0 — Manage Supply Chain for Physical Products

Human involvement H% at each of five HITL maturity levels · L3 Process → L2 Process Group → L1 Domain · Equal-weighted aggregation · Pattern types derived using the HITL Derivation Method

Maturity Level	M1 Undesigned	M2 Emerging	M3 Defined	M4 Optimized	M5 Leading
Description	No intentional HITL design. AI tools exist but aren't systematically applied.	Tools selectively deployed. AI awareness growing. No systematic design yet.	Intentional HITL design applied. Pattern types classified. Decision rights documented.	AI mature across all patterns. Continuous improvement. Benchmarked against industry.	Maximum AI absorption with principled human oversight. Future state north star.

H% by Pattern Type per Maturity Level

Pattern	Nature of Work	M1	M2	M3	M4	M5
Decision	Judgment & Authority	92%	88%	80%	72%	65%
Knowledge	Synthesis & Interpretation	85%	78%	65%	52%	42%
Document	Content Generation	72%	58%	40%	28%	18%
Transaction	Rules-Based Processing	60%	48%	35%	22%	12%
Exception	Non-Standard Resolution	95%	92%	82%	75%	68%

The Human-AI Partnership Framework · HITL Maturity Ratios

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PCF	Process / Process Group		Pattern	M1 Undesigned	M2 Emerging	M3 Defined	M4 Optimized	M5 Leading
4.0	Manage Supply Chain for Physical Products L1 Domain Aggregate			72%	63%	50%	38%	28%
4.1	Plan for and Align Supply Chain Resources (8 processes) L2 Avg			83%	76%	63%	51%	41%
4.1.1	↳ Develop production and materials strategies	Knowledge		85%	78%	65%	52%	42%
4.1.2	↳ Manage demand for products	Knowledge		85%	78%	65%	52%	42%
4.1.3	↳ Create materials plan	Knowledge		85%	78%	65%	52%	42%
4.1.4	↳ Create and manage master production schedule	Transaction		60%	48%	35%	22%	12%
4.1.5	↳ Plan distribution requirements	Knowledge		85%	78%	65%	52%	42%
4.1.6	↳ Establish distribution planning constraints	Decision		92%	88%	80%	72%	65%
4.1.7	↳ Review distribution planning policies	Knowledge		85%	78%	65%	52%	42%
4.1.8	↳ Develop quality standards and procedures	Knowledge		85%	78%	65%	52%	42%
4.2	Procure Materials and Services (5 processes) L2 Avg			81%	74%	62%	50%	41%
4.2.1	↳ Provide sourcing governance	Knowledge		85%	78%	65%	52%	42%
4.2.2	↳ Develop category plan	Knowledge		85%	78%	65%	52%	42%
4.2.3	↳ Select suppliers and develop/maintain contracts	Decision		92%	88%	80%	72%	65%
4.2.4	↳ Order materials and services	Transaction		60%	48%	35%	22%	12%
4.2.5	↳ Manage suppliers	Knowledge		85%	78%	65%	52%	42%
4.3	Produce/Assemble/Test Product (4 processes) L2 Avg			60%	48%	35%	22%	12%
4.3.1	↳ Schedule production	Transaction		60%	48%	35%	22%	12%
4.3.2	↳ Produce/Assemble product	Transaction		60%	48%	35%	22%	12%
4.3.3	↳ Perform quality testing	Transaction		60%	48%	35%	22%	12%
4.3.4	↳ Maintain production records and manage lot traceability	Transaction		60%	48%	35%	22%	12%
4.4	Manage Logistics and Warehousing (5 processes) L2 Avg			65%	54%	41%	28%	18%
4.4.1	↳ Provide logistics governance	Knowledge		85%	78%	65%	52%	42%
4.4.2	↳ Plan and manage inbound material flow	Transaction		60%	48%	35%	22%	12%
4.4.3	↳ Operate warehousing	Transaction		60%	48%	35%	22%	12%
4.4.4	↳ Operate outbound transportation	Transaction		60%	48%	35%	22%	12%
4.4.5	↳ Deliver last mile delivery services	Transaction		60%	48%	35%	22%	12%

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Pattern types: Decision=Judgment & Authority · Knowledge=Synthesis & Interpretation · Document=Content Generation · Transaction=Rules-Based Processing · Exception=Non-Standard Resolution · M3 (Defined) = HITL design intent baseline